

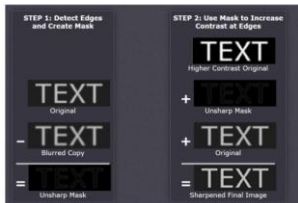
23. Perform Sharpening of Image using unsharp masking.

Unsharp masking

$$f_s(x, y) = f(x, y) - \tilde{f}(x, y)$$

sharpened image = original image – blurred image

- to subtract a blurred version of an image produces sharpening output image.



AIM:

Sharpening of Image using unsharp masking.

PROGRAM:

```
from PIL import Image, ImageFilter
```

```
image_path = r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png"
```

```
try:
```

```
    # Open the image    im1 =  
    Image.open(image_path)
```

```
    # Apply Unsharp Mask filter  
    im2 = im1.filter(  
        ImageFilter.UnsharpMask(radius=3, percent=200, threshold=5)  
    )
```

```
    # Save the output image    im2.save(r"C:\Users\Susmitha\Downloads\ITA0510  
    Lab\Unsharp_Mask_Image.png")
```

```
    # Display the original and sharpened images  
    im1.show(title="Original Image")    im2.show(title="Unsharp  
    Mask Image")
```

```
except FileNotFoundError:
```

```
    print("Error: Image not found!")
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.